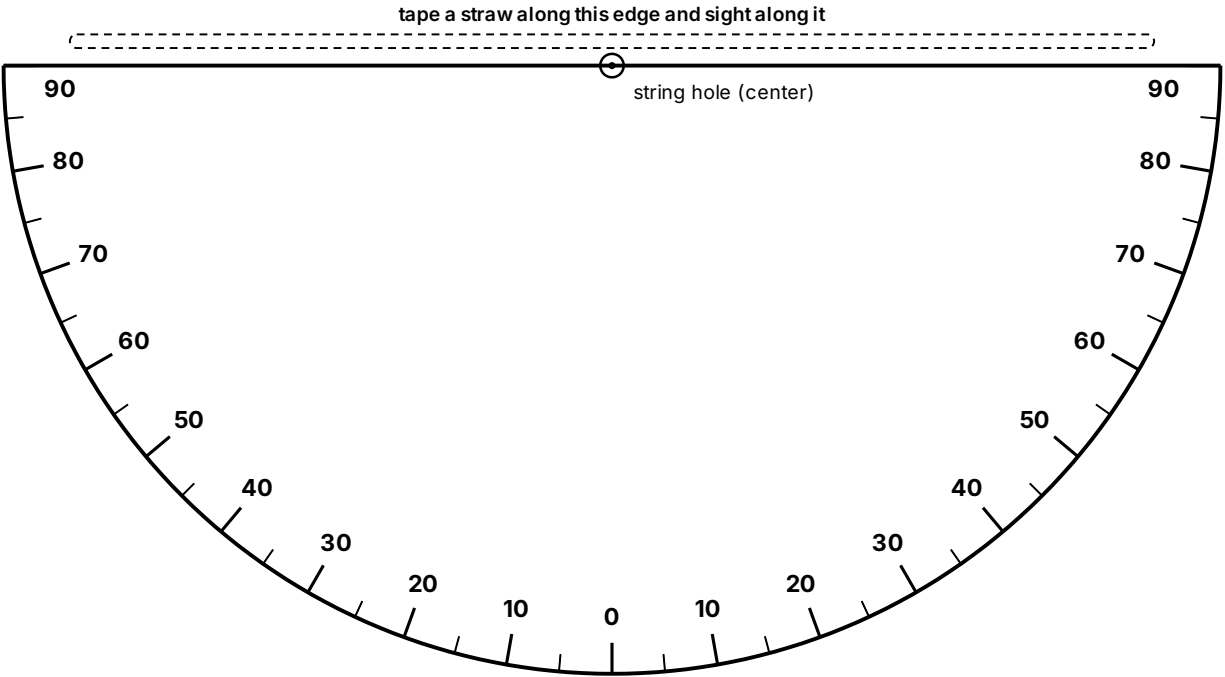


Paper clinometer



Cut along the flat top edge and the curved outline.

ASSEMBLE

- 01 Cut out the half circle along the curved outline.
- 02 Tape a straight straw along the top straight edge.
- 03 Punch the center dot, thread a string through it, and tie a washer or paper clip on the end as a weight.
- 04 Sight the treetop through the straw, let the string hang free, then pinch it against the scale and read the number.
- 05 Self-check: sight something level (reads 0) and straight up (reads 90).

FIND THE TREE HEIGHT

Tree height = eye height + (distance to the tree × tan of the angle). Use the standoff distance the instructor marked.

ANGLE	TAN
10	0.18
20	0.36
30	0.58
40	0.84
45	1.00
50	1.19
60	1.73
70	2.75

Example: eye height 1.5 m, distance 10 m, angle 40° gives 1.5 + 10 × 0.84 = about 9.9 m.

Team route card

From Trees to Tech • TTT-02 Forest Sensor Sprint • Team Route Card

Team _____ Date _____

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

CHECKPOINT	VISIT ORDER	TEMP (°F)	HUMIDITY (%)	LIGHT (RELATIVE)	SOIL MOISTURE (1 TO 10)	NOTES
Reference (calibrate)						
A						
B						
C						
D						
E						

Our recommended sensor site: _____

Why (use your data): _____

The checkpoints are the spots the instructor marked outside.

cut here

From Trees to Tech • TTT-02 Forest Sensor Sprint • Team Route Card

Team _____ Date _____

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

CHECKPOINT	VISIT ORDER	TEMP (°F)	HUMIDITY (%)	LIGHT (RELATIVE)	SOIL MOISTURE (1 TO 10)	NOTES
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